

Math Class 9th Federal Board Chapter

04 Factorization Full Test

2 MCQs -
✓ 2 Short Ans.
✓ 1 Long Qs!

16-18 marks

Factorization	[SLO: M-09-A-17]: Identify common factors, trinomial factoring, concretely, pictorially and symbolically.	Formative	K	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.
	[SLO: M-09-A-18]: Factorize quadratic and cubic algebraic expressions. $a^4 + a^2b^2 + b^4$ or $a^4 + b^4$ ✓ $x^2 + px + q$ $ax^2 + bx + c$ $(ax^2 + bx + c)(ax^2 + bx + d) + k$ $(x + a)(x + b)(x + c) + k$ ✓ $(x + a)(x + b)(x + c)(x + d) + kx^2$ $a^3 + 3a^2b + 3ab^2 + b^3$ ✓ $a^3 - 3a^2b + 3ab^2 - b^3$ ✓ $a^3 \pm b^3$	Summative	U	Question(s) will be asked in the annual examination
	[SLO: M-09-A-19]: Find highest common factor and least common multiple of algebraic expressions and know relationship of LCM and HCF.	Summative	U/K	Question(s) will be asked in the annual examination
	[SLO: M-09-A-20]: Find square root of algebraic expression by factorization and division.	Summative	U/K	Question(s) will be asked in the annual examination
	[SLO: M-09-A-21]: Apply the concepts of factorization of quadratic and cubic algebraic expressions to real world problems (such as engineering, physics, and finance.)	Summative	A	Question(s) will be asked in the annual examination

$ax^2 + bx + c$
Trinomial

$ax + b$
Binomial
 $ax \rightarrow$ Monomial

SECTION A - OBJECTIVE

15 Marks

Q.1. Choose the correct option. Each MCQ carries 1 mark.

1. For two polynomials A and B, $HCF \times LCM =$

(A) $A + B$

(B) $A - B$

(C) $(HCF)^2$

(D) $A \times B$ ✓

$$A \times B = HCF \times LCM$$

$$HCF = \frac{A \times B}{LCM}$$

2. Which is the highest common factor of $-12x^2y^2$, $6xy^3$, $24x^2y^2$?

X (A) $-6xy$

HCF?

(C) $6xy^2$ ✓

X (B) $6x^2y^3$

X (D) $-24x^2y^3$

$-12, 6, 24$ ✓

x^2, x^1, x^2

y^2, y^3, y^2

$$y^3 = y \cdot y^2$$

$$\frac{y^3}{y^2} = y$$

3. What is the least common multiple of $7x - 6xy$ and $5xy^3 - 3x^2$?

X (A) $(7 - 6y)(5x^3 - 3x)$

X (B) $(7x - 6xy)(5y^3x - 3x^2)$

X (C) $x(7 - 6y)$

(D) $x(7 - 6y)(5y^3 - 3x)$ ✓

LCM?

$$x(7 - 6y), x(5y^3 - 3x)$$

$$\underline{x(7-6y)}, \quad \underline{x(5y^3-3x)}$$

$$\boxed{LCM = x(7-6y)(5y^3-3x)}$$

4. Factorization of $4a^2 - 12ab + 9b^2$ is:

☒ (A) $(2a - 3b)^2$

☒ (C) $(4a + 3b)^2$

☒ (B) $(2a + 3b)^2$

☒ (D) $(4a - 3b)^2$

$$(2a)^2$$

$$(3b)^2$$

$$\sqrt{4a^2} = 2a$$

$$\begin{aligned} \sqrt{(a-b)^2} &= a^2 + b^2 - 2ab \\ \sqrt{(a+b)^2} &= a^2 + b^2 + 2ab \\ \sqrt{(-a-b)^2} &= a^2 + b^2 + 2ab \end{aligned}$$

5. HCF of $3x^2$ and $6x^3$ is:

(A) $3x$

☒ (C) $3x^2$

☒ (B) $6x^2$

(D) $3x^3$

$$\underline{3, 6}$$

$$(3)$$

$$(3)(2)$$

$$x^2 \cdot x = x^3$$

$$\boxed{HCF = 3 \times x^2}$$

6. Factorization of $49a^2 - 28ab + 4b^2$ is:

☒ (A) $(7a + 2b)^2$

☒ (C) $(7a - 2b)^2$

☒ (B) $(49a - 4b)^2$

☒ (D) $(7a - 28b)^2$

$$(7a)^2 (2b)^2$$

$$49a^2 + 4b^2 - 2(7a)(2b)$$

7. What is the square root of $\left[-\left(2x + \frac{1}{x} + 1\right)\right]^2$?

(A) $\pm \left(2x + \frac{1}{x} + 1\right)^2$

(B) $\left(2x + \frac{1}{x} + 1\right)^2$

(C) $2x + \frac{1}{x} + 1$ ✓

(D) $\sqrt{2x + \frac{1}{x} + 1}$

$(-1)^2 = +1$ ✓ $\pm \sqrt{\left(2x + \frac{1}{x} + 1\right)}$ ✓
 $= \pm \left(2x + \frac{1}{x} + 1\right)$ ✓

8. If the product of HCF and LCM of two polynomials is $(x^3 - y^3)(x + y)$, then what will be the product of these polynomials?

(A) $x^4 - y^4$

(B) $(x - y)(x^2 + y^2)$

(C) $(x^2 - y^2)(x^2 + xy + y^2)$ ✓

(D) $(x^2 - y^2)(x^2 - xy + y^2)$

$P \times Q = (x^3 - y^3)(x + y)$
 $= (x - y)(x^2 + xy + y^2)(x + y)$
 $= (x^2 - y^2)(x^2 + xy + y^2)$

9. What is the square root of $(15x^2 - 7y^2)^4$?

(A) $\pm(15x^2 - 7y^2)^2$ ✓

(B) $\pm(15x^2 - 7y^2)$

(C) $\pm(15x - 7y)^2$ ✗

(D) $\pm(15x - 7y)$

$\sqrt{(15x^2 - 7y^2)^4} \Rightarrow 4 \times \frac{1}{2} = 2$

10. What is the difference of $\frac{-3x}{x + y}$ and $\frac{x}{x + y}$?

(A) $\frac{-2x}{x + y}$

(B) $\frac{-3x - x}{2x + 2y}$

(C) $\frac{3x^2}{x + y}$

(D) $\frac{-4x}{x + y}$ ✓

$\frac{-3x - x}{x + y}, \quad \frac{x - (-3x)}{x + y}$

$$\begin{array}{r} -3-1 \\ \hline -4 \end{array} \quad , \quad \begin{array}{r} -5x-10 \\ \hline x+2 \end{array} \quad , \quad \begin{array}{r} x-(-5) \\ \hline x+5 \end{array}$$

$$\begin{array}{r} -3x-x \\ \hline -4x-4 \\ \hline x+2 \end{array}$$

11. Factors of $-2 - a + a^2$ are: ✓

(A) $(a-2)(a-1)$

(B) $(a+1)(a+2)$

(C) $(a+2)(a-1)$

(D) $(a+1)(a-2)$ ✓

✓ Use Calculator

$$ax^2 + bx + c = 0$$

$$\begin{array}{l} a \leq 2 \\ (a-2) \\ a \leq -1 \\ (a+1) \end{array}$$

$$a^2 - a - 2$$

$$a=1, b=-1, c=-2$$

$$\begin{array}{r} -2a+a-2 \\ \hline a^2-a-2 \end{array}$$

12. Factorization of $x^2 - x + \frac{1}{4}$ is:

(A) $\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)$

(B) $\left(x + \frac{1}{2}\right)(x-1)$

(C) $\left(x - \frac{1}{2}\right)(x+1)$

(D) $\left(x - \frac{1}{2}\right)\left(x - \frac{1}{2}\right)$ ✓

$$\left(x - \frac{1}{2}\right)^2 = \left(x - \frac{1}{2}\right)\left(x - \frac{1}{2}\right)$$

$$\left(x - \frac{1}{2}\right)^2 = \left(x - \frac{1}{2}\right)\left(x - \frac{1}{2}\right)$$

13. If the product of two polynomials is $(a-b)^2(a^2+ab+b^2)$ and their HCF is $a-b$, what is their LCM?

(A) $a^3 - b^3$

(B) $(a-b)^3(a^2+ab+b^2)$

(C) $(a-b)^2(a^2+ab+b^2)$

(D) a^2+ab+b^2

$$HCF \times LCM = (a-b)^2(a^2+ab+b^2)$$

$$LCM = \frac{(a-b)^2(a^2+ab+b^2)}{(a-b)}$$

$$= a^3 - b^3 \quad \checkmark$$

$$= (a^2 + ab + b^2)(a^2 - ab + b^2)$$

Ans

Factorize the expression $8x^4 - 26x^2m^2 + 18m^4$.

$$\begin{aligned}
 &= 2(4x^4 - 13x^2m^2 + 9m^4) \\
 &= 2\left(\underbrace{(2x^2)^2 + (3m^2)^2}_{\substack{\uparrow \quad \uparrow}} + 2(2x^2)(3m^2) - 13x^2m^2\right) \\
 &= 2\left((2x^2 + 3m^2)^2 - 12x^2m^2 - 13x^2m^2\right) \\
 &= 2\left((2x^2 + 3m^2)^2 - 25x^2m^2\right) \\
 &= 2\left((2x^2 + 3m^2)^2 - (5xm)^2\right) \\
 &= 2(2x^2 + 3m^2 + 5xm)(2x^2 + 3m^2 - 5xm)
 \end{aligned}$$

Ans

$$2((a^2 - b^2))^2 = 2(a - b)(a + b)$$

Factorize

(b) $a^4 + 64b^4$

$$\begin{aligned}
&= (a^2)^2 + (8b^2)^2 + 2(a^2)(8b^2) - 2(a^2)(8b^2) \\
&= (a^2 + 8b^2)^2 - 16a^2b^2 \\
&= (a^2 + 8b^2)^2 - (4ab)^2 \\
&= (a^2 + 8b^2 + 4ab)(a^2 + 8b^2 - 4ab)
\end{aligned}$$

A

$$(x + a)(x + b)(x + c) + k$$

Factorize

$$(x + 1)(x + 5)(x + 6) + 12$$

Let

$$P(x) = (x + 1)(x^2 + 6x + 5x + 30) + 12$$

$$= (x + 1)(x^2 + 11x + 30) + 12$$

$$= x^3 + 11x^2 + 30x + x^2 + 11x + 30 + 12$$

$$= x^3 + 12x^2 + 41x + 42$$

Let $p = 42$, $q = 1$ (leading coeff)

The possible rational roots are

$$\pm 1, \pm 2, \pm 3, \pm 6, \pm 7, \pm 12, \pm 42$$

1st p.m.

$$\frac{p}{q} = \pm 1, \pm 2, \pm 3, \pm 6, \pm 7, \pm 12, \pm 42$$

Let try $x = -2$ so
put in above expr.

$P(-2) = 0$, so, $x = -2$
is the root of above expression

so, $(x+2)$ is a factor of $P(x)$

$$\begin{array}{r} x^2 + 10x + 21 \\ \text{Now } x+2 \overline{) x^3 + 12x^2 + 41x + 42} \\ \underline{-(x^2 + 2x^2)} \\ 10x^2 + 41x + 42 \\ \underline{-(10x^2 + 20x)} \\ 21x + 42 \\ \underline{-(21x + 42)} \\ 0 \end{array}$$

As Quotient $Q(x) = x^2 + 10x + 21$

$$Q(x) = x^2 + 3x + 7x + 21$$

$$Q(x) = x^2 + 5x + 6$$

$$= x(x+3) + 7(x+3)$$

$$Q(x) = (x+3)(x+7)$$

Hence

$$(x+1)(x+5)(x+6) + 12 = (x+2)(x+3)(x+7)$$

Type-IV

Factorizing expressions of the forms:

$$(a) \quad a^3 + 3a^2b + 3ab^2 + b^3 = (a+b)^3$$

$$(b) \quad a^3 - 3a^2b + 3ab^2 - b^3 = (a-b)^3$$

Example 7: Factorize the following expressions.

(i) $27a^3 + 1 + 27a^2 + 9a$

(ii) $x^3 + \frac{3}{x} - \frac{1}{x^3} - 3x$

(iii) $4x^2(2x-15) - 25(5-6x)$

$$= 27a^3 + 27a^2 + 9a + 1$$

$$(i) = (3a)^3 + 3(3a)^2(1) + 3(3a)(1)^2 + (1)^3$$

$$= (3a+1)^3$$

(ii) $x^3 + \frac{3}{x} - \frac{1}{x^3} - 3x$

$$\Rightarrow \frac{a^3 - 3a^2b + 3ab^2 - b^3}{(\because a=x, b=\frac{1}{x})}$$

$$= x^3 - 3x + \frac{3}{x} - \frac{1}{x^3}$$

(iii)

$$= 8x^3 - 60x^2 - 125 + 150x$$

$$= 8x^3 - 60x^2 + 150x - 125$$

$$= (2x)^3 - 3(2x)^2(5) + 3(2x)(5) - (5)^3$$

$$= (2x-5)^3 \quad \Delta$$

Type-V

(a) $a^3 + b^3$ (b) $a^3 - b^3$

We have studied in the previous unit that:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2) \quad \text{and} \quad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Example 8: Factorize: $x^3 + 27y^3$

$$= (x)^3 + (3y)^3$$

$$\begin{aligned}
 &= (x + 3y)(x^2 - (x)(3y) + (3y)^2) \\
 &= (x + 3y)(x^2 - 3xy + 9y^2)
 \end{aligned}$$

Example 9: Factorize $x^6 - y^6$.

$$\begin{aligned}
 &= (x^3)^2 - (y^3)^2 \\
 &= (x^3 - y^3)(x^3 + y^3) \\
 &= (x - y)(x^2 + xy + y^2)(x + y)(x^2 - xy + y^2) \\
 &= (x - y)(x + y)(x^2 + xy + y^2)(x^2 - xy + y^2)
 \end{aligned}$$

Example 10: Factorize $8a^3 - 125b^3 - 2a + 5b$

$$\begin{aligned}
 &= (2a)^3 - (5b)^3 - 2a + 5b \\
 &= (2a - 5b)((2a)^2 + (2a)(5b) + (5b)^2) - 2a + 5b \\
 &= (2a - 5b)(4a^2 + 10ab + 25b^2) - (2a - 5b)
 \end{aligned}$$

$$= (2a - 5b)(4a^2 + 10ab + 25b^2 - 1)$$

$$P = -\frac{1}{2} \Rightarrow 2P = -1, (2P + 1)$$

Q.2. Attempt all parts. Each part carries 4 marks.

(i) Simplify:

$$\frac{5}{5 + p - 18p^2} - \frac{2}{2 + 5p + 2p^2}$$

$$P = \frac{5}{9} \Rightarrow 9P = 5 \quad (5 - 9P)$$

Sol:

Let

$$5 + p - 18p^2$$

$$= (5 - 9P)(2P + 1)$$

$$= 5 + 10P - 9P - 18P^2$$

$$= 10P + 5 - 18P^2 - 9P$$

$$= 5(1 + 2P) - 9P(1 + 2P)$$

$$= 5 + 10P - 9P - 18P^2$$

$$= (1 + 2P)(5 - 9P)$$

calculator use

$$\text{Also } 2 + 5p + 2p^2$$

$$= 2 + p + 4p + 2p^2$$

$$= 1(2 + p) + 2p(2 + p)$$

$$= (2 + p)(1 + 2p)$$

$$P = -\frac{1}{2}$$

$$2P = -1$$

$$(2P + 1)$$

$$P = -2$$

$$(P + 2)$$

$$(P + 2)(2P + 1)$$

$$2P^2 + P + 4P + 2$$

$$\text{Sol, } \frac{5}{5+p-18p^2} - \frac{2}{2+5p+2p^2}$$

$$= \frac{5}{(1+2p)(5-9p)} - \frac{2}{(2+p)(1+2p)}$$

$$= \frac{5(2+p) - 2(5-9p)}{(1+2p)(2+p)(5-9p)}$$

$$= \frac{\cancel{10} + 5p - \cancel{10} + 18p}{(1+2p)(2+p)(5-9p)}$$

$$= \frac{23p}{(1+2p)(2+p)(5-9p)}$$

(ii) Find the HCF of the polynomials

$$x^3 + 2x^2 - 4x - 8 \quad \text{and} \quad 2x^3 + 7x^2 + 4x - 4.$$

Sol: By division method

$$\begin{array}{r} 2 \\ x^3 + 2x^2 - 4x - 8 \overline{) 2x^3 + 7x^2 + 4x - 4} \\ \underline{2x^3 + 4x^2 - 8x - 16} \\ 3x^2 + 12x + 12 \end{array}$$

$$\sqrt{\begin{matrix} 2x^3 & +4x^2 & -8x & -16 \\ \ominus & \ominus & \oplus & \oplus \end{matrix}}$$

Remainder $3x^2 + 12x + 12$
 Divide by 3 $\Rightarrow x^2 + 4x + 4$ ✓
 $x-2$

$$\begin{array}{r} \sqrt{x^2+4x+4} \overline{) \begin{matrix} x^3+2x^2-4x-8 \\ x^3+4x^2+4x \\ \hline -2x^2-8x-8 \\ -2x^2-8x-8 \\ \hline 0 \end{matrix}} \end{array}$$

Thus, HCF = $x^2 + 4x + 4$ ✓
 $= (x+2)^2$ ✓
 $-12 \times 3 = -36$ ✓ $18x^2$ ✓

(iii) Factorize the quadratic expression:

$$3x^2 + 16x - 12.$$

$$\begin{aligned} &= 3x^2 - 2x + 18x - 12 \\ &= x(3x-2) + 6(3x-2) \\ &= (x+6)(3x-2) \end{aligned}$$

↓
 $-25-4 = -29$

$$\begin{aligned} x &= \frac{2}{3} \Rightarrow 3x = 2 \\ (3x-2) \\ x &= -6 \Rightarrow (x+6) \\ (x+6)(3x-2) \\ x(3x-2) + 6(3x-2) \\ 3x^2 - 2x + 18x - 12 \end{aligned}$$

$$-25-4 = \boxed{-29}$$

$$3x^2 - 2x + 18x - 12$$

Example: $10z^2 - 29z + 10$

$$(p=100) \\ (s=-29)$$

$$= 10z^2 - 4z - 25z + 10$$

$$2 \times 50$$

$$-4 \times -25$$

$$= 2z(\underline{5z-2}) - 5(\underline{5z-2})$$

$$= (2z-5)(\underline{5z-2})$$

$$(\underline{2z-5})(\underline{5z-2})$$

$$z = \frac{5}{2}$$

$$(2z-5)$$

$$2z(5z-2)$$

$$-5(5z-2)$$

$$z = \frac{2}{5}$$

$$(5z-2)$$

$$10z^2 - 4z - 25z + 10$$

Example $6 - 7x - 5x^2$

$$= 6 + 3x - 10x - 5x^2$$

$$2 \times 15$$

$$+3 \times 10$$

$$(6)(-5)$$

$$P = \boxed{-30x^2}$$

$$s = -7$$

$$5x = 3 \Rightarrow (3-5x)$$

$$R = 5x^2$$

$$x = \frac{3}{5}, x = -2$$

$$(\underline{3-5x})(\underline{x+2})$$

$$(3-5x)$$

$$3(x+2) - 5x(x+2)$$

$$3x + 6 - 5x^2 - 10x$$

$$6 + 3x - 10x - 5x^2$$

$$1 \dots 3, 11$$

$$= -5x^2 - 7x + 6$$

$$= - (5x^2 + 7x - 6)$$

$$= - (5x^2 + 7x - 6)$$

$$= - (5x^2 + 10x - 3x - 6)$$

$$= - (5x(x+2) - 3(x+2))$$

$$= - (5x - 3)(x+2)$$

$$= (3 - 5x)(x+2)$$

Ex 2

$$8 + 6x - 5x^2$$

$$= 8 + 10x - 4x - 5x^2$$

$$= 2(4 + 5x) - x(4 + 5x)$$

$$= (2 - x)(4 + 5x)$$

Ex 3

$$p = (2x - 3) = -63$$

Ex 4

$$2x^2 - 5xy - 3y^2$$

$$= 2x^2 - 6xy + xy - 3y^2$$

$$= 2x(x - 3y) + y(x - 3y)$$

$$x = 3/5$$

$$(5x-3)(x+2)$$

$$5x(x+2) - 3(x+2)$$

$$5x^2 + 10x - 3x - 6$$

$$p = -40x^2$$

$$s = 6x$$

$$(-4x \times 10x)$$

$$x = 2$$

$$(2-x)$$

$$x = -4/5$$

$$(5x+4)$$

$$(2-x)(4+5x)$$

$$2(4+5x) - x(4+5x)$$

$$8 + 10x - 4x - 5x^2$$

$$x = 3y$$

$$(x-3y)$$

$$\begin{aligned}
 &= 2x(x-3y) + y(x-3y) \\
 &= (2x+y)(x-3y)
 \end{aligned}$$

$x = -\frac{y}{2}$
 $(2x+y)$
 $(x-3y)(2x+y)$
 $2x^2 + xy - 6xy - 3y^2$
 $x(2x+y) - 3y(2x+y)$

(iv) For the algebraic expressions $P = 6x^2y$ and $Q = 9xy^2$, show that

Sol: $6 = 2 \times \underline{3}, \quad 9 = \underline{3}^2$

$$P \times Q = \text{LCM} \times \text{HCF}.$$

$$\text{LCM} = 3^2 \times 2 \times x^2 \times y^2$$

$$\text{LCM} = 18x^2y^2$$

$$\text{HCF} = 3xy$$

$$\begin{array}{r|l}
 2 & 6-9 \\
 \hline
 3 & 3-9 \\
 \hline
 3 & 1-3 \\
 \hline
 & 1-1
 \end{array}$$

$$\begin{aligned}
 \text{L.H.S} &= P \times Q \\
 &= (6x^2y)(9xy^2) \\
 &= 54x^3y^3
 \end{aligned}$$

$$1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1$$

$$\begin{aligned}
 R.H.S &= LCM \times HCF \\
 &= 18x^2y^2 \times 3xy \\
 &= 54x^3y^3
 \end{aligned}$$

$$\Rightarrow L.H.S = R.H.S$$

$$\text{Hence } P \times Q = LCM \times HCF$$

proved

(v) Factorize:

$$(\underline{x^2 - 7x + 4})(\underline{x^2 - 7x + 5}) - 2.$$

Sol:

$$= \text{Let } y = x^2 - 7x$$

$$\Rightarrow = (y+4)(y+5) - 2$$

$$= y^2 + 5y + 4y + 20 - 2$$

$$= y^2 + 9y + 18$$

$$= y^2 + 3y + 6y + 18$$

6x3

$$= y + 3y + 6y + 18$$

$$= y(y+3) + 6(y+3)$$

$$= (y+6)(y+3)$$

-6-1

$$= (x^2 - 7x + 6)(x^2 - 7x + 3)$$

$$= (x^2 - x - 6x + 6)(x^2 - 7x + 3)$$

$$= x(x-1) - 6(x-1)(x^2 - 7x + 3)$$

$$= (x-1)(x-6)(x^2 - 7x + 3)$$

(vi) If $A = 4a^2b$ and $B = 6ab^2$, verify that

$$\underline{2^2} \underline{2 \times 3}$$

$$A \times B = \text{LCM} \times \text{HCF}.$$

Sol:

$$\text{LCM} = 2^2 \times 3 \times a^2 \times b^2$$

$$\boxed{\begin{array}{l} \text{LCM} = 12a^2b^2 \\ \text{HCF} = 2ab \end{array}}$$

$$\text{L.H.S} = A \times B$$

$$= 24a^3b^3$$

$$\text{R.H.S} = 24a^3b^3 \Rightarrow$$

L.H.S = R.H.S
Proved

$$R.H.S = 24ab \quad \text{Proves}$$

(vii) Factorize:

$$x^3 + x^2 - 4x - 4.$$

$$\begin{aligned}
 &= x^2(x+1) - 4(x+1) \\
 &= (x^2 - 4)(x+1) \\
 &= (x^2 - 2^2)(x+1) \\
 &= (x-2)(x+2)(x+1) \\
 &= (x-2)(x+1)(x+2) \quad \checkmark
 \end{aligned}$$

Ex-3

(viii) Find the HCF and LCM of

$$x^2 - 9 \quad \text{and} \quad x^2 - x - 6.$$

Solⁿ

$$x^2 - 9 = (x)^2 - (3)^2$$

$$x^2 - 9 = (x-3)(x+3) \quad \checkmark$$

Now

$$x^2 - x - 6 = (x+2)(x-3) \quad \checkmark$$

$$= x^2 + 2x - 3x - 6$$

$$= x(x+2) - 3(x+2) \quad \checkmark$$

$$\therefore \dots (x-2)(x+2)(x+3)$$

$$LCM = (x-3)(x+2)(x+3)$$

$$HCF = x-3$$

(ix) Find the square root of

$$4x^4 + 12x^3 + 29x^2 + 30x + 25.$$

$$\frac{12x^3}{4x^4} = 3x$$

$$2x^2 + 3x + 5$$

$2x^2$	$4x^4 + 12x^3 + 29x^2 + 30x + 25$ $\underline{4x^4}$ $\hline$
$4x^2 + 3x$	$12x^3 + 29x^2 + 30x + 25$ $\underline{12x^3 + 9x^2}$ $\hline$
$4x^2 + 6x + 5$	$20x^2 + 30x + 25$ $\underline{20x^2 + 30x + 25}$ $\hline$
	0

Thus square root is

Ans

$$= \pm (2x^2 + 3x + 5)$$

Example 27: Find the square root of $\left(a + \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right)$.

$$= a^2 + \frac{1}{a^2} + 2 \cdot (a) \cdot \left(\frac{1}{a}\right) - 4\left(a - \frac{1}{a}\right)$$

$$= a^2 + \frac{1}{a^2} + 2 - 4\left(a - \frac{1}{a}\right)$$

$$= a^2 + \frac{1}{a^2} - \underline{\underline{2}} + 2 + 2 - 4\left(a - \frac{1}{a}\right)$$

$$= \left(a - \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right) + 4$$

$$= \left(a - \frac{1}{a}\right)^2 - 2\left(a - \frac{1}{a}\right)(2) + (2)^2$$

$$= \left(a - \frac{1}{a} - 2\right)^2$$

$$\sqrt{\left(a + \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right)} = \pm \sqrt{\left(a - \frac{1}{a} - 2\right)^2}$$

$$= \boxed{\pm \left(a - \frac{1}{a} - 2\right)}$$

Ans.

Ans

Example 27: Find the square root of $\left(a + \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right)$.

Method $\swarrow$

$$= a^2 + \frac{1}{a^2} + 2 - 4\left(a - \frac{1}{a}\right) \rightarrow \textcircled{1}$$

Let $\boxed{x = a - \frac{1}{a}}$

$$\Rightarrow x^2 = a^2 + \frac{1}{a^2} - 2$$

$$\Rightarrow \boxed{x^2 + 2 = a^2 + \frac{1}{a^2}}$$

So, Eqn. $\textcircled{1}$ becomes

$$\left(a + \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right) = (x^2 + 2) + 2 - 4x$$

$$= x^2 - 4x + 4$$

$$\left(a + \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right) = (x - 2)^2$$

taking sq. root on both sides

Taking Sqr. root on D.N., we get

$$\sqrt{\left(a + \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right)} = \pm \sqrt{(x-2)^2}$$

$$= \pm (x-2)$$

$$= \pm \left(a - \frac{1}{a} - 2\right)$$

Ans

Example 27: Find the square root of $\left(a + \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right)$.

$$= \frac{a^2 + 1 + 2 - 4a + \frac{4}{a}}{a^2}$$

$$= a^2 - 4a + 2 + \frac{4}{a} + \frac{1}{a^2}$$

$$\frac{-4a + 2}{2a}$$

By using Division Method.

$$a - 2 - \frac{1}{a}$$

$$\frac{-7 + \left(\frac{1}{a}\right)}{2a}$$

$$\begin{array}{r}
 a \quad \begin{array}{l} a^2 - 4a + 2 + \frac{4}{a} + \frac{1}{a^2} \\ \hline a^2 \\ \hline \end{array} \\
 2a - 2 \quad \begin{array}{l} -4a + 2 + \frac{4}{a} + \frac{1}{a^2} \\ \hline -4a + 4 \\ \hline \end{array} \\
 2a - 4 - \frac{1}{a} \quad \begin{array}{l} -2 + \frac{4}{a} + \frac{1}{a^2} \\ \hline -2 + \frac{4}{a} + \frac{1}{a^2} \\ \hline \end{array} \\
 \hline
 0
 \end{array}$$

Thus

$$\sqrt{\left(a + \frac{1}{a}\right)^2 - 4\left(a - \frac{1}{a}\right)} = \pm \left(a - \frac{1}{a} - 2\right)$$

Example 28: Find the square root of $(2a + 1)(2a + 3)(2a + 5)(2a + 7) + 16$.

=

$$(2a + 1)(2a + 7)(2a + 3)(2a + 5) + 16$$

$$= (4a^2 + 14a + 2a + 7)(4a^2 + 10a + 6a + 15) + 16$$

$$= (\underline{4a^2 + 16a + 7})(\underline{4a^2 + 16a + 15}) + 16$$

$$\text{Let } \boxed{x = 4a^2 + 16a}$$

$$= (x + 7)(x + 15) + 16$$

$$= x^2 + 15x + 7x + 105 + 16$$

$$= x^2 + 22x + 121$$

$$= x^2 + 11x + 11x + 121$$

$$= x(x + 11) + 11(x + 11)$$

$$= (x + 11)(x + 11)$$

$$= (x + 11)^2$$

$$= (4a^2 + 16a + 11)^2$$

$$\sqrt{(\quad)(\quad)(\quad)(\quad)+16} = \pm \sqrt{(4a^2+16a+11)^2}$$

$$= \pm (4a^2+16a+11)$$

As

Example 5b: Factorize the expression

$$(z+1)(z-5)(z-9)(z-3)+44$$

$$\begin{cases} 1-9=-8 \\ -5-3=-8 \end{cases}$$

$$= (z+1)(z-9)(z-5)(z-3)+44$$

$$= (\underline{z^2-8z-9})(\underline{z^2-8z+15})+44$$

$$\text{Let } z^2-8z = x$$

$$= (x-9)(x+15)+44$$

$$\begin{pmatrix} 4 \\ 135 \end{pmatrix}$$

$$= x^2+15x-9x-135+44$$

$$= x^2+6x-91$$

$$= x^2+13x-7x-91$$

$$= x^2 + 13x - 7x - 91$$

$$= x(x+13) - 7(x+13)$$

$$= (x-7)(x+13)$$

$(-1)(-1)$
 $p=7$
 $\checkmark$

$$= (z^2 - 8z - 7)(z^2 - 8z + 13)$$

$\checkmark$ $\checkmark$

(x) Factorize:

$$(x-2)(x-6)(x-3)(x-4) - 2x^2.$$

$(-2)(-6) = 12 \checkmark$
 $(-3)(-4) = +12 \checkmark$ *Rough Work*

$(x+1)(x+2)(x+3)(x+6) - 3x^2$

$(x+1)(x+6) \quad (x+2)(x+3) - 3x^2$

(x) Factorize:

$$(x-2)(x-6)(x-3)(x-4) - 2x^2.$$

$$= (x^2 - 8x + 12)(x^2 - 7x + 12) - 2x^2$$

$$1 \cdot 2 \dots 6 \cdot 3 / x^2 - 8x + 12 - 7x + 12 - 2x^2$$

$$= \underbrace{(x^2 + 12 - 8x)}_{\text{let } y = x^2 + 12} (\underbrace{x^2 + 12 - 7x}_{y - 7x}) - 2x^2$$

$$\text{let } y = x^2 + 12$$

$$= (y - 8x)(y - 7x) - 2x^2$$

$$= y^2 - 7xy - 8xy + 56x^2 - 2x^2$$

$$= 54x^2 - 15xy + y^2$$

$$= y^2 - 15xy + 54x^2$$

$$= y^2 - 6xy - 9xy + 54x^2$$

$$= y(y - 6x) - 9x(y - 6x)$$

$$= (y - 6x)(y - 9x)$$

$$= (x^2 + 12 - 6x)(x^2 + 12 - 9x)$$

$$= (x^2 - 6x + 12)(x^2 - 9x + 12)$$

Handwritten signature

SECTION C - LONG QUESTIONS

Q.3. Attempt all parts. Each part carries 8 marks. Show complete working.

(i) To convert

$$x^4 + 4x^3 + 10x^2 + 10x + 5$$

into a perfect square:

- What should be added to it?
- What should be subtracted from it?

Handwritten long division for completing the square:

$$\begin{array}{r}
 x^2 + 2x + 3 \\
 \hline
 x^4 + 4x^3 + 10x^2 + 10x + 5 \\
 \underline{-(x^4)} \\
 4x^3 + 10x^2 + 10x + 5 \\
 \underline{-(4x^3 + 4x^2)} \\
 6x^2 + 10x + 5 \\
 \underline{-(6x^2 + 12x + 9)} \\
 -2x - 4
 \end{array}$$

Labels on the left side of the division:

- x^2 (next to x^4)
- $2x^2 + 2x$ (next to $4x^3 + 10x^2$)
- $2x^2 + 4x + 3$ (next to $6x^2 + 10x + 5$)

$$| \quad \boxed{-2x - 4}$$

Remainder = $-2x - 4$

a) We add $2x + 4$ ✓

b) we subtract $-2x - 4$

$$\begin{aligned} & -2x - 4 - (-2x - 4) \\ & = -2x - 4 + 2x + 4 = 0 \end{aligned}$$

c) What should be the value of x ?

$$-2x - 4 = 0$$

$$-2x = 4$$

$$\frac{-25x^2}{2x^2} = (-12.5)$$

$$|x = -2|$$

$$\frac{6}{2x^2} = \frac{3}{x^2}$$

(ii) Find the values of p and q if

$$x^4 - 12x^3 + px + q$$

is a complete square.

$$x^2 - 6x - 18$$

is a complete square.

Sol: $x^2 - 6x - 18$

$$\begin{array}{r}
 x^4 - 12x^3 + px + 9 \\
 \underline{-(2x^4 - 12x^3 + 6x^2)} \\
 6x^2 + px + 9 \\
 \underline{-(6x^2 - 36x + 36)} \\
 36x + 9 - 36 \\
 36x - 27
 \end{array}$$

For a perfect square.

Remainder = 0

$$36x - 27 = 0$$

$$(36)x + (-27) = 0x + 0$$

$$\underline{\underline{(P-216) \times \dots}} \quad \uparrow \quad \underline{\underline{(V - \dots)}}$$

$$\Rightarrow P-216 = 0$$

$$\boxed{P=216}$$

$$A180, \quad 9-224 = 0$$

$$\boxed{P=324}$$

f

$$2 \times \left(\frac{(K-4)}{2y^2} \right)$$

48/28

(iii) For what value of k does the expression

$$y^4 + 4y^2 + k + \frac{8}{y^2} + \frac{4}{y^4}$$

become a perfect square?

$$y^2 + 2 + \frac{K-4}{2y^2}$$

$$y^2$$

$$\begin{array}{r} y^4 + 4y^2 + k + \frac{8}{y^2} + \frac{4}{y^4} \\ \underline{y^4} \end{array}$$

$$2y^2 + 2$$

$$\begin{array}{r} 4y^2 + k + \frac{8}{y^2} + \frac{4}{y^4} \\ \underline{4y^2 + 4} \end{array}$$

$$k-4 + \frac{8}{y^2} + \frac{4}{y^4}$$

$$2y^2 + 4 + \frac{k-4}{2y^2}$$

$$\frac{k-4}{y^2} + \frac{8}{y^2} + \frac{4}{y^4}$$

$$\frac{k-4}{y^2} + \frac{2(k-4)}{y^2} + \frac{(k-4)^2}{(2y^2)^2}$$

$$\frac{8}{y^2} - \frac{2(k-4)}{y^2} + \frac{4}{y^4} - \frac{(k-4)^2}{4y^4}$$

For a perfect square, Remainder = 0

$$\Rightarrow \frac{8}{y^2} - \frac{2(k-4)}{y^2} - \frac{(k-4)^2}{4y^4} + \frac{4}{y^4} = 0$$

$$\frac{1}{y^2} (8 - 2k + 8) + \left(4 - \frac{(k-4)^2}{4}\right) \frac{1}{y^4} = 0$$

$$\Rightarrow (16-2k) \times \frac{1}{y^2} + \left(\frac{16-(k-4)^2}{4} \right) \times \frac{1}{y^4} = \frac{0}{y^2} + \frac{0}{y^4}$$

By comparing

$$16 - 2K = 0$$

$$2k = 16$$

$$16 - (k-4)^2 = 0$$

$$\sqrt{(k-4)^2} = \pm \sqrt{16}$$

$$2k = 16$$

$$\boxed{k = 8}$$

$$\sqrt{(k-4)^2} = \sqrt{16}$$

$$k-4 = \pm 4$$

$$k-4 = 4 \quad | \quad k-4 = -4$$

$$\boxed{k = 8} \quad | \quad \boxed{k = 0}$$

Thus $\boxed{k = 8}$ ✓ Any

(iv) Find the HCF and LCM in simplified form using the factorization method:

$$2x^3 - x^2 - 8x + 4 \quad \text{and} \quad 2x^3 + 3x^2 - 8x + 3.$$

$$2x^3 - x^2 - 8x + 4$$

$$= x^2(\underline{2x-1}) - 4(\underline{2x-1})$$

$$= (2x-1)(x^2-4)$$

$$\boxed{2x^3 - x^2 - 8x + 4 = (2x-1)(x-2)(x+2) \rightarrow \textcircled{1}}$$

N/m

$$\downarrow$$

$$2x^3 + 3x^2 - 8x + 3$$

$$\quad \quad \quad +3x^2 \quad \quad$$

$$= 2x^3 - 2x^2 + 2x^{2+3x^2} - 8x + 3$$

$$= 2x^2(x-1) + 5x^2 - 8x + 3$$

$$= 2x^2(x-1) + 5x^2 - 5x - 3x + 3$$

$$= 2x^2(x-1) + 5x(x-1) - 3(x-1)$$

$$= (x-1)(\underline{2x^2 + 5x - 3}) \quad +6x1$$

$$= (x-1)(2x^2 + 6x - x - 3)$$

$$= (x-1)(2x(x+3) - 1(x+3))$$

$$\boxed{2x^3 + x^2 - 8x + 3 = (x-1)(x+3)(2x-1)} \quad \text{--- (2)}$$

$$\boxed{\text{HCF} = 2x-1}$$



$$\boxed{\text{LCM} = (2x-1)(x-1)(x+3)(x-2)(x+2)}$$

1/14

$$Q(x) = 2x^3 + 3x^2 - 8x + 3$$

Let $p = 3, q = 2$

All possible rational roots are

$$\frac{p}{q} = \pm 1, \pm 3, \pm \frac{3}{2}, \pm \frac{1}{2} \quad \begin{matrix} p = \pm 1, \pm 3 \\ q = (\pm 1) \pm 2 \end{matrix}$$

Let try $x = 1 \Rightarrow (x-1)$

By checking, $x = 1$ is the root of $2x^3 + 3x^2 - 8x + 3$
 $\Rightarrow Q(1) = 0$

So, $(x-1)$ is the factor of $Q(x)$

$$2x^2 + 5x - 3$$

$$(x-1) \overline{) 2x^3 + 3x^2 - 8x + 3}$$

$$\begin{array}{r}
 \underbrace{x-1} \overline{) 2x^3 + 3x^2 - 8x + 3} \\
 \underline{2x^3 \quad - 2x^2} \\
 5x^2 - 8x + 3 \\
 \underline{5x^2 \quad - 5x} \\
 -3x + 3 \\
 \underline{-3x + 3} \\
 0
 \end{array}$$

As Quotient $q(x) = (2x^2 + 5x - 3)$

$$\begin{aligned}
 2x^2 + 5x - 3 &= 2x^2 + 6x - x - 3 \\
 &= 2x(x+3) - 1(x+3) \\
 &= (x+3)(2x-1)
 \end{aligned}$$

$$\begin{array}{l}
 \swarrow \\
 x = -3 \\
 (x+3) \\
 \swarrow \\
 x = \frac{1}{2} \\
 (2x-1) \\
 (2x-1)(x+3) \\
 \underline{2x^2 + 6x - x - 3}
 \end{array}$$

$$(a+b)^2 = a^2 + b^2 + 2(a)(b)$$

$$\begin{aligned}
 \underline{(a+b)^3} &= a^3 + b^3 + 3(a)(b)(a+b) \\
 &= \underline{a^3 + b^3 + 3a^2b + 3ab^2}
 \end{aligned}$$

a

$$\begin{aligned}
 (a-b)^3 &= a^3 - b^3 - 3(a)(b)(a-b) \\
 &= \boxed{a^3 - b^3 - 3a^2b + 3ab^2}
 \end{aligned}$$